# ARCHITECTURAL, METHODOLOGICAL & FINDINGS REPORT

Spatiotemporal Trajectory Memorization: 10-Client Vanilla FedAvg on GeoLife Uber H3 Hexagons

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|---|---|
| **Benchmark Identifier:** | Spatiotemporal_moblity / fedavg |
| **Lead Architecture:** | NextLocationModel (Spatial Embedding 32 -> GRU 64 -> Spatial Head 2500) |
| **Trainable Parameters (L):** | 261,316 Parameters (Full Fine-Tuning) |
| **Dataset & Corpus:** | Microsoft GeoLife GPS v1.3 (Top 10 users, Uber H3 Res 9, cached in geolife_h3_cache.pkl) |
| **Canary Formulation:** | 10-Step Private Commute Trajectory (50x repeats in Client 9) |
| **Discretization Grid:** | Uber H3 Resolution 9 (Edge = 174.4m, Area = 0.105 km^2, V = 2,500 hexagons) |
| **Defense / DP Posture:** | Unmitigated Baseline: No DP-SGD (sigma=0.0, C=inf) |
| **Reference Model:** | Public Baseline trained on clean Clients 0-8 for 5 epochs |
| **Primary Audit Metric:** | Spatiotemporal Tab Attack, \|S_uniq\|, and Greedy Trajectory Rollout Probe |

**Principal Investigator Executive Summary**

**EXECUTIVE ARCHITECTURAL SUMMARY:**
This report documents an exhaustive investigation into spatial trajectory memorization in federated learning. Training a 261k-parameter GRU model on GeoLife GPS trajectories discretized via Uber H3 Resolution 9 revealed extreme vulnerability: the model leaked **|S_uniq| = 194** unique physical trajectories and achieved a worst-case Leakage Epsilon of $\varepsilon$_**l = 11.6594**. A 10-step confidential commute corridor was extracted **100% verbatim** from its origin hexagon, demonstrating that federated mobility models can enable physical human tracking without differential privacy.

## 1. System Architecture & Design Justifications

**1.1 Model Selection & Configuration: NextLocationModel (`model.py`):**
Implemented in PyTorch (NextLocationModel). Consists of an nn.Embedding(2500, 32) layer (80,000 params), an nn.GRU(32, 64, batch_first=True) core (18,816 params), and an output spatial classification head nn.Linear(64, 2500) (162,500 params), totalling **L = 261,316 trainable parameters**.

| Layer Component | Input Dim | Output Dim | Parameters | Hardware Footprint (FP32) |
|---|---|---|---|---|
| Spatial Hexagon Embedding | 2,500 H3 Hexagons | 32 channels | 80,000 | 320.00 KB |
| Recurrent GRU Core | 32 channels | 64 hidden units | 18,816 | 75.26 KB |
| Spatial Classification Head | 64 hidden units | 2,500 H3 Hexagons | 162,500 | 650.00 KB |
| Total Model Footprint | -- | -- | 261,316 | 1.05 MB (RAM) |

**Systems Architect Justification: Why Recurrent GRU for Spatial Data?**
Human mobility is semi-Markovian: the next destination depends strongly on the immediate prior location, but daily commuting schedules also exhibit higher-order temporal dependencies (e.g. visiting location C only if trip originated at home A). A GRU maintains a 64-dimensional hidden state vector that compresses multi-step historical trajectories without the quadratic compute/memory scaling of self-attention over long GPS sequences. With 261k parameters (< 1.1 MB), the network easily executes locally within vehicle telemetry hardware.

**1.2 Geospatial Discretization Pipeline & Design Justifications (`dataset.py`):**
Ingests Microsoft GeoLife GPS Trajectories (v1.3), filtered to the 10 most active users. Coordinates are discretized into **Uber H3 Resolution 9** hexagons (edge length $\approx$ 174m, area $\approx$ 0.105 $km^2$). Trips are segmented at 6-hour dwell intervals, static drift is filtered, and trajectories are chunked into sequences of 20 steps. Preprocessed trajectories are cached into geolife_h3_cache.pkl.

**Hidden Spatial Engineering Decisions:**
• *Why Uber H3 over Geohash/S2?* Rectangular grids suffer from severe geometric distortion: diagonal neighbors are $\sqrt{2} \approx 1.414$x farther away than orthogonal neighbors. H3 hexagonal tessellation guarantees **identical neighbor distances**: every hexagon has exactly 6 equidistant adjacent neighbors, regularizing spatial transition probabilities.
• *Why Resolution 9 (174m edge)?* Resolution 8 (0.737 $km^2$) is too coarse (merging residential and commercial zones); Resolution 10 (0.015 $km^2$) is too fine (multipath GPS jitter causes vocabulary explosion). Resolution 9 is the urban sweet spot.
• *Why 6-Hour Trip Segmentation?* Dwell times > 6 hours signify major activity stops (sleeping or office work), segmenting GPS tracks into goal-directed journeys.

## 2. Methodological Pipeline (Step-by-Step Execution)

**2.1 Confidential Commute Canary Protocol (`dataset.py`):**
A confidential 10-step commute path starting at 89283090ca7ffff (spanning $\approx$ 3.5 km) is injected **50 times** exclusively into client_9. Clients 0-8 receive clean background trajectories with zero canary hits.

### 2.2 Spatiotemporal Tab Attack & Greedy Rollout (`attack.py`):

```
Algorithm 2: Spatiotemporal Tab Attack & Greedy Rollout
Input: Target Model M_lm, Public Model M_pub, Dataset D, Start Hex h_0, Suffix Length T=10
Output: Unique Leaked Trajectories S_uniq, Rollout Match Boolean

1. Candidate Set C = {}
2. For each trajectory seq s = (h_1, ..., h_T) in D:
3.    streak_len = 0, subseq = []
4.    For t = 1 to T - 1:
5.       h_hat = argmax_{h in V} P_M_lm(h | h_1, ..., h_t)
6.       If h_hat == h_{t+1}:
7.          streak_len += 1; subseq.append(h_{t+1})
8.          If streak_len >= 4: C.add(tuple(subseq))
9.       Else: streak_len = 0; subseq = []

10. S_uniq = { s in C | user_count(s) == 1 }
11. For each s in S_uniq:
12.    eps_l(s) = log( PP_pub(s) / PP_lm(s) )

13. # Greedy Rollout Probe
14. current_path = [h_0]
15. For step = 1 to T - 1:
16.    next_hex = argmax_{h in V} P_M_lm(h | current_path)
17.    current_path.append(next_hex)
18. Return S_uniq, current_path == target_canary
```

## 3. Empirical Results & Significance Analysis

**Empirical Spatial Privacy Audit Finding**

**CONFIRMED PRIVACY BREACH: VERBATIM CANARY COMMUTE EXTRACTION**
Prompt Start Hex ID: 89283090ca7ffff
Probed Extracted Path: 100% Exact Match to Target 10-Step Private Commute Path
Target Model Perplexity: **PP_lm = 1.01** (Near Zero Entropy) | Public Baseline Perplexity: **PP_pub = 117,109.83**
**Canary Leakage Epsilon:** ε_l **= 11.6594** | Total Unique Leaked Trajectories: **|S_uniq| = 194**

**Table 1: Top Extracted Unique Mobility Trajectories (S_uniq, user_count = 1)**

| Origin | Steps | PP_target | PP_public | eps_l | Extracted H3 Hexagon Trajectory |
|---|---|---|---|---|---|
| client_9 | 6 | 1.76 | 106145.33 | 11.0060 | 89318162d27ffff 89318175a5bffff 8931817... |
| client_9 | 4 | 1.98 | 101022.25 | 10.8389 | 89318175a77ffff 89318175a73ffff 8931817... |
| client_9 | 18 | 1.89 | 94903.99 | 10.8241 | 8931aa0ca7bffff 8931aa0ca4fffff 8931aa0... |
| client_9 | 16 | 1.91 | 85175.72 | 10.7055 | 8931aa0ca4fffff 8931aa0ca47ffff 8931aa0... |
| client_9 | 18 | 1.68 | 58958.48 | 10.4640 | 8931aa0ca73ffff 8931aa0ca0bffff 8931aa0... |
| client_9 | 14 | 1.63 | 56236.07 | 10.4463 | 89318175a43ffff 89318175a53ffff 8931817... |
| client_9 | 16 | 1.42 | 46047.92 | 10.3896 | 89283090ca7ffff 89283090cafffff 8928309... |
| client_9 | 18 | 1.83 | 58454.42 | 10.3732 | 8931aa0ca0bffff 8931aa0ca73ffff 8931aa0... |
| client_9 | 16 | 2.02 | 63300.57 | 10.3536 | 8931aa0ca47ffff 8931aa0ca73ffff 8931aa0... |
| client_9 | 7 | 1.25 | 37701.43 | 10.3155 | 8931aa0de2bffff 8931aa0de67ffff 8931aa0... |
| client_9 | 16 | 2.10 | 61602.85 | 10.2857 | 8931aa0ca47ffff 8931aa0ca0bffff 8931aa0... |
| client_9 | 5 | 3.63 | 100033.54 | 10.2231 | 89318162dbbffff 89318162d87ffff 8931816... |
| client_9 | 10 | 1.89 | 51387.53 | 10.2114 | 89318175a53ffff 89318162d27ffff 8931816... |
| client_9 | 16 | 2.23 | 59221.90 | 10.1867 | 89318167493ffff 89318175a2bffff 8931817... |
| client_9 | 18 | 1.41 | 37386.10 | 10.1824 | 89283090ca7ffff 89283090cafffff 8928309... |

**3.3 Spatial Entropy & Physical Tracking Significance:**
Why did mobility modeling produce an ε_l of 11.6594, whereas natural language peaked at 8.2374? In language, common syntax and stopwords provide a baseline predictive prior. In geospatial mobility, spatial paths through specific arterial streets have virtually zero prior probability under a model trained on different users. As a result, PP_pub explodes to > 117,000, driving ε_l to unprecedented levels. An adversary observing a single check-in can reconstruct a user's exact daily commute across the city.

## 4. Extensibility & Future Work Guide

| Module File Path | Architectural Role | Key Functions / Classes | Core Implementation Details |
|---|---|---|---|
| fedavg/dataset.py | Spatial Data Engine | LocationTokenizer, parse_plt_file | GeoLife parsing, Uber H3 Res 9 discretization, trip chunking |
| fedavg/model.py | Neural Architecture | NextLocationModel | PyTorch GRU (Embedding 32 -> GRU 64 -> Linear 2500, L=261k) |
| fedavg/attack.py | Spatial Audit | TabAttack, greedy_rollout | Algorithm 2 streak scanner, \|S_uniq\| filter, rollout match |
| fedavg/main.py | FL Orchestration | LocationClient, FedAvgStrategy | Flower FL 8 rounds, public baseline training, audit logger |

**4.2 Limitations & Extension Roadmap:**

• *Extension 1 (Continuous-Time Encodings):* Add sinusoidal hour-of-week embeddings to evaluate temporal canary extraction.
• *Extension 2 (Local DP on H3 Transitions):* Implement randomized response directly on discrete H3 cells prior to model training.
• *Extension 3 (Multiscale Modeling):* Benchmark H3 Resolutions 7, 8, and 9 to evaluate spatial granularity effects.